

Predicting Bagrut Success

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1 Introduction

Every year Israeli high-school students sit the *Bagrut*, the national matriculation examinations that gate university admission. A long-standing question in educational policy is how much a student’s outcome is fixed by the socioeconomic status of the *municipality* their school sits in, versus what the *school itself* does with its resources. This project answers that question with a reproducible, cross-validated machine-learning pipeline.

The main question is whether municipal socioeconomic status and school-level institutional resources, *combined*, meaningfully predict Israeli high-school Bagrut achievement and advanced-track participation. Two secondary questions follow. The resilience question: which subject (Mathematics or English) is more *resilient* to socioeconomic and institutional disparity, and do schools exist that outperform their peers despite limited resources? The school-factors question: which specific school-level factors (budget allocation, class size, sector, or supervision type) most strongly predict outcomes, *independent* of municipal wealth?

We model four school-level targets: the average Bagrut grade and the advanced (5-unit) participation rate, each for Mathematics and English. A municipal score cannot capture what happens inside a specific school, and separating the two is what makes the policy question answerable.

Revisions since the interim stage. The study began as a municipal-only design. Following review we added the Ministry’s school-level dataset and rebuilt the pipeline from ingestion onward. Later revisions introduced cross-validation grouped by school, per-target feature selection, explicit handling of privacy-suppressed grades, a municipal-versus-school ablation, and a nested protocol in which feature selection and tuning run inside training folds only.

Code availability. All data-processing and modelling code, with per-stage documentation, is openly available at <https://github.com/Yousef-Shihade/predicting-bagrut-success>. Every figure and number below traces to a file in that repository.

2 Data Description and Collection

This study integrates three independent, publicly available sources, each describing Israeli high schools from a different angle. **(1) Bagrut exam records** (2013–2016), released under the Israeli Freedom of Information Law and mirrored on Kaggle, give per-subject results for each school and year. **(2) The Central Bureau of Statistics (CBS) socioeconomic index** rates every locality on a 1–10 cluster capturing municipal wealth, education, and employment. **(3) The Ministry of Education’s school budget and institutional report** provides per-institution funding, staffing, class size, sector, and supervision type. Source 1 supplies the *outcomes*, source 2 the *municipal* environment, and source 3 lets us look *inside* the school itself. The unit of analysis is the *school-year*, not the student.

Structure and storage. The Bagrut file is a UTF-8 CSV. The CBS and Ministry files are Excel workbooks with awkward layouts (a header offset, Hebrew column names, and a malformed style block requiring a lenient reader). Each source is ingested by a dedicated loader and cached as a clean CSV (Table 1).

Table 1: The three integrated data sources: one supplies the outcomes, one the municipal context, and one the school’s own institutional profile. This split is what lets the study separate “where a school is” from “what a school does”.

Dataset	Source	Rows	Cols	Grain (key)	Coverage
Bagrut exam records	Israeli FOI (Kaggle)	69,638	9	subject cell (1,063 schools, 315 cities)	2013–2016
CBS socioeconomic index	Israel CBS	1,208	7	locality (cluster 1–10)	2015 index
Ministry budget & institution report	Israel Min. of Education	4,718	26	school (semel)	2014/15 snapshot

Linking the sources. The tables are joined on *two different keys*, because **semel** is not shared consistently: in the Bagrut file it is a *school* code, in the CBS file a *locality* code, giving zero overlap. We therefore (i) join Bagrut to CBS on a *normalised Hebrew locality name* through a four-stage cascade (exact $\rightarrow$ structural $\rightarrow$ hand-verified crosswalk $\rightarrow$ conservative fuzzy match), integrating **99.44%** of records, and (ii) join Bagrut to the Ministry report directly on **semel** (a shared *school* code), integrating **99.68%**. A left join can multiply rows when the right-hand key repeats, so each lookup table is de-duplicated on its join key first. This matters because the normalised locality name is not unique in the CBS file. Row preservation was verified rather than assumed: the merged table has exactly 69,638 rows, matching the Bagrut input, and **98.7%** of records carry a complete profile from all three sources.

Biases and challenges. Three were material. The key incompatibility above is resolved by the name cascade rather than a fragile single-pass match. The Ministry report is a single-year (2014/15) snapshot broadcast across all four Bagrut years, a documented limitation that assumes school resourcing is reasonably stable over 2013–2016. Finally, the unmatched 0.56% of records (boarding and regional schools with no single home municipality) are *flagged, not forced*, avoiding an invented socioeconomic value.

3 Data Cleaning and Preprocessing

Cleaning addressed four concerns: the missing target, predictor imputation, outliers, and encoding.

The missing target is not missing at random. 21.4% of Bagrut grade values are absent, but the absence is structured: suppressed cells have a median of 7 test takers against 28 where the grade is present. This is the Ministry protecting the privacy of small cohorts, not random loss. We therefore *never impute the target*. School-level grades are aggregated over observed cells only, so no outcome value is invented.

Imputation is measured, not assumed. Rather than assuming multivariate imputation by chained equations (MICE) recovers missing predictor values well, we test it directly: we hide 8% of a fully observed feature (the CBS socioeconomic score), reconstruct it with MICE, and score the reconstruction against the withheld truth over 25 independent masks. MICE beats a median baseline on all 25 runs, but the margin depends on what else is available. With the municipal **cluster** in the predictor set MICE reaches $R^2 = 0.954 \pm 0.006$, a number that flatters the method because **cluster** is a discretisation of the same index ($r = 0.97$). Withholding that near-duplicate and the outcome columns, MICE still reaches $R^2 = 0.331 \pm 0.056$ (RMSE 0.755) while the median baseline never rises above zero, and it wins all 25 masks in every predictor set tested. MICE reconstructs each gap from the other observed columns, whereas the median

returns one value everywhere and cannot track variation at all. Per-variant results are in the repository.

Modelling itself uses complete cases. The masking study is a diagnostic. Imputed values are *not* carried into the modelling table. Rows missing any numeric predictor are dropped, costing 9.0% to 12.5% of the target-present rows depending on the outcome, leaving 2,963 to 3,226 school-years per target. Missing categorical values enter as an all-zero indicator pattern.

Outliers are flagged, and kept in the primary analysis. Two complementary detectors, Isolation Forest (global anomalies) and the Local Outlier Factor (local-density anomalies), were run on a nine-feature space. They agree on only 49 records (Jaccard 0.17), confirming they capture genuinely different notions of “anomalous”. That feature space includes outcome variables, so excluding the flagged rows would condition the evaluation sample on the target and could quietly remove legitimate high or low performers. The primary results therefore retain every valid school. Repeating the evaluation with the 49 records dropped changes mean R^2 by +0.001, with per-target shifts between -0.023 and $+0.013$, well inside the fold-to-fold standard deviation. The choice is immaterial to the findings.

Encoding and scaling. Categorical attributes (sector, supervision, district, legal status, education stage, locality type) are one-hot encoded, with Hebrew values mapped to English first for readable outputs. The exam **year** is encoded as four independent periods rather than a numeric trend, since with only four years there is no basis to assume an equal step between them. Linear models are standardised inside the cross-validation folds, never on the full dataset, so no test-fold information leaks into the scaling. Trees need no scaling.

4 Machine Learning Models

Every outcome here is a number on a continuous scale, not a label, so this is a regression problem throughout. Four school-level outcomes are modelled, each on its own: two grade averages on a 0–100 scale (**math_avg_grade**, **english_avg_grade**) and two advanced-track (5-unit) participation rates on a 0–1 scale (**math_5unit_participation**, **english_5unit_participation**).

Four model families compete on equal terms. Rather than commit to one algorithm, we span the useful complexity range from a regularised linear fit to modern boosted trees (Table 2). Every family is tuned within the same inner folds using a predefined model-specific search space and up to 25 candidate configurations. Ridge’s seven regularisation values are evaluated exhaustively. An earlier version of this study tuned only the boosted-tree model and left the others at defaults, which makes any conclusion about a winner an artefact of unequal effort rather than a property of the data.

Table 2: The four competing model families. All are tuned under the same nested protocol using model-specific hyperparameter spaces.

Model	Family	Why it fits this problem
Ridge	Linear (L2)	Stable linear baseline the trees must beat. L2 shrinkage handles the many one-hot dummies well.
SGD linear	Linear (SGD)	Second linear fit by stochastic gradient descent, a check that the linear result is not a solver artefact.
Random Forest	Bagged trees	Captures non-linearities and interactions without scaling. Bagging resists overfitting.
HistGradient-Boosting	Boosted trees	Efficient boosted-tree model capturing nonlinear relationships and feature interactions in encoded tabular data.

Nested cross-validation keeps the evaluation honest. Two kinds of leakage matter. Structurally, each school contributes up to four year-rows (2013–2016), so folds are formed by school using

GroupKFold(`seml`). More subtly, if feature selection or tuning sees the whole dataset then the evaluation folds have already influenced the model and scores become optimistic. We therefore *nest*: an outer GroupKFold supplies untouched evaluation folds, and within each outer *training* fold only we run VIF pruning, Boruta selection, and an inner GroupKFold search. Every number in Table 3 is an outer-fold result, reported as a mean and standard deviation across folds.

Table 3: Comparative model performance: outer-fold R^2 (mean $\pm$ standard deviation over five folds), each family tuned under the same protocol. Random Forest leads on every target, though the families sit within roughly one standard deviation of each other on the grade outcomes.

Target	Ridge	SGD	Rand. Forest	HistGB
math_avg_grade	0.405 $\pm$ 0.066	0.402 $\pm$ 0.058	0.420$\pm$0.050	0.414 $\pm$ 0.054
english_avg_grade	0.410 $\pm$ 0.045	0.410 $\pm$ 0.043	0.452$\pm$0.042	0.441 $\pm$ 0.043
math_5unit_participation	0.323 $\pm$ 0.043	0.323 $\pm$ 0.041	0.395$\pm$0.037	0.378 $\pm$ 0.030
english_5unit_participation	0.502 $\pm$ 0.036	0.504 $\pm$ 0.037	0.557$\pm$0.053	0.539 $\pm$ 0.064

Random Forest is selected for every target, though two observations temper the ranking. On the grade outcomes all four families lie within about one standard deviation of one another, so the algorithm matters far less than the features. The trees separate more clearly on participation, where the outcome is bounded and the relationships less linear. Feature selection was stable across folds: 11–12 features for the Math targets and 13–15 for English.

5 Feature Selection Methods

Feature selection runs in two stages: first remove predictors that are redundant with each other, then keep only those that carry real signal for the outcome.

Removing collinearity by iterative VIF pruning. From 15 numeric candidates, instead of dropping every feature above the variance-inflation threshold ($VIF > 5$) at once, we drop the single worst offender and recompute until every survivor clears it, because inflation is usually shared between a pair. Several features look collinear at first (the teaching-budget ratio at $VIF \approx 77$, individual-instruction hours at ≈ 58 , `cluster` at ≈ 20), but most of that collapses once the true culprit goes. Only three drops are needed: two redundant budget ratios and the raw `index_value`, which duplicates the `cluster` built from it. `cluster` itself survives once `index_value` is gone, leaving twelve numeric features.

Keeping relevance by Boruta, per target. On those 12 survivors plus seven one-hot categorical blocks (sector, supervision, district, legal status, education stage, locality type, year), totalling 49 encoded columns, Boruta runs separately for each target. Using a 200-tree random forest it compares every real feature against randomly shuffled shadow copies over up to 100 iterations, retaining one only when its importance repeatedly exceeds the 90th percentile of the shadow importances. A predictor survives only if its signal is dependably stronger than noise.

The result is a nine-feature core confirmed for all four targets: locality population, the nurture index, average class size, school size, and five per-student budget ratios (tuition, perimeter, projects, purchases, transport). Each target adds its own on top (Table 4). Notably `cluster` is *not* in that core: municipal status is confirmed for three targets but rejected outright for English average grade. Five core features are budget lines, and every target confirms at least one sector, supervision, or district attribute, a first sign that school-level structure carries signal the municipal cluster does not. No `year` dummy is confirmed anywhere, which says nothing about the shape of a temporal trend but does mean no single exam year carried enough independent signal to be retained.

Feature selection is repeated independently inside every outer training fold for performance evaluation. Table 4 reports the final descriptive Boruta selection obtained by refitting on all available modelling rows after evaluation. This final fit is used for interpretation and SHAP analysis, not for estimating predictive performance.

Table 4: Boruta selections per target: the shared nine-feature core plus each target’s own confirmed additions. Extras are genuinely target-specific, not cumulative. Municipal `cluster` appears as an addition for three targets and is rejected for `english_avg_grade`.

Target	Additions beyond the nine-feature core	Total
Math grade	<code>cluster</code> , <code>supervision_Haredi</code>	11
Math 5-unit	<code>cluster</code> , <code>district_North</code> , <code>sector_Jewish</code>	12
English 5-unit	<code>cluster</code> , <code>private_hours_per_student</code> , <code>sector_Jewish</code> , <code>supervision_Haredi</code>	13
English grade	<code>district_North</code> , <code>private_hours_per_student</code> , <code>special_</code> <code>ed_share</code> , <code>sector_Bedouin</code>	13

6 Performance Evaluation and Cross-Validation

Three metrics, each guarding a different failure. Every model is scored on three complementary regression metrics rather than one. R^2 reports the share of outcome variance explained, which is scale-free and so comparable across the four targets. RMSE gives the typical error in the target’s own units and penalises large misses heavily. MAE gives the plain average absolute error, which is robust to a few extreme cases. Reporting all three means no single number can flatter a model: a low RMSE paired with a poor R^2 , for instance, would expose a model that looks precise only because the outcome barely varies. Because the outcomes are continuous, class imbalance does not apply, but the target distributions were still inspected. Both participation outcomes are right-skewed, advanced Math especially, and that skew reappears in the residuals below.

Errors in interpretable units. For the selected model the grade targets carry an MAE of 4.11 points (Math) and 3.45 (English) on the 0–100 scale, with RMSE 5.37 and 4.53. The participation targets carry an MAE of 5.8 and 13.1 percentage points, with RMSE 8.5 and 17.8. The English participation error is the largest in absolute terms, consistent with that outcome having the widest spread across schools.

Residual diagnostics. Figure 1 shows the distribution of the nested out-of-fold residual (actual minus predicted) per target. Because these predictions come from outer folds never used for selection or tuning, the diagnostic matches the protocol behind Table 3. Every mean residual is within 0.03 of zero, so there is no material *global* bias, though this does not rule out subgroup bias or calibration error at the extremes. The grade models are near-symmetric. The participation models are centred on zero but mildly right-skewed, because the outcome is itself bounded and right-skewed: the model slightly over-predicts the many low-participation schools and under-predicts the rare high ones. No panel shows a second peak or heavy structured tail, though residual-versus-fitted and subgroup diagnostics would be needed before attributing the remainder to irreducible noise.

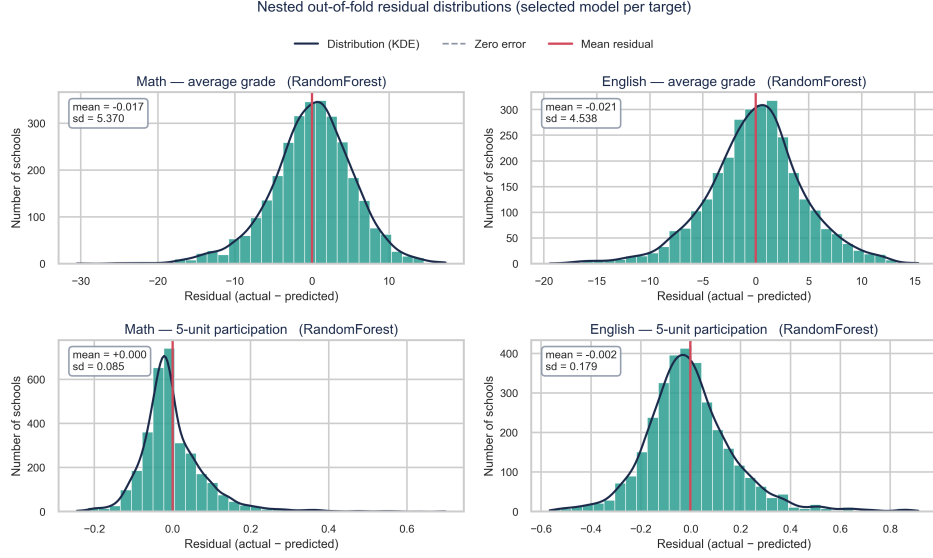


Figure 1: Nested out-of-fold residual distributions for the selected model, with each panel’s mean and standard deviation annotated. The coral mean line sits on the grey zero line in all four panels.

7 Feature Analysis and Importance

Two analyses answer the main research question. SHAP attributes each prediction to individual features, and a controlled ablation measures what the school-level data adds over municipal socioeconomics alone.

What the models actually rely on. Figure 2 ranks the nine features confirmed for all four targets, plus municipal `cluster` for comparison, by mean absolute SHAP value as a share of each target’s total so the outcomes sit on one scale. These values come from a RandomForest refitted on all rows using the same Boruta-selected features per target, matching the family Table 3 selects, so they explain that model rather than held-out predictions, and they report magnitude, not direction.

The `nurture_quintile` leads every target, taking 19.7% to 36.2% of total attribution. This is the Ministry’s nurture (*tipuach*) index recorded *per school* on a 1–5 scale where higher means more disadvantaged: mean combined grade falls monotonically from 85.3 at quintile 1 to 74.1 at quintile 5. It is missing for 11.9% of school-years, which leave the complete-case sample. Municipal `cluster` never exceeds 5.3% and is dropped entirely for English average grade. Both measure socioeconomic standing, so this is not a finding that socioeconomics do not matter but one about *granularity*: schools inside one municipality differ enough that the town-level average conceals most of the effect.

Beyond that index the leading features are institutional. The transport budget line ranks second for every target, but needs care: it is negative for 86% of institutions in the raw Ministry file and 83% in the complete-case modelling sample, indicating a net accounting adjustment rather than gross spending, so its meaning needs confirmation against the Ministry’s data dictionary. Class size and school size follow, and school size is especially strong for advanced Math participation (16.5%), consistent with a large cohort being what makes opening a 5-unit class viable.

Feature importance per target (SHAP, 9-feature shared core plus municipal cluster)
navy = school-level nurture index, red = municipal cluster

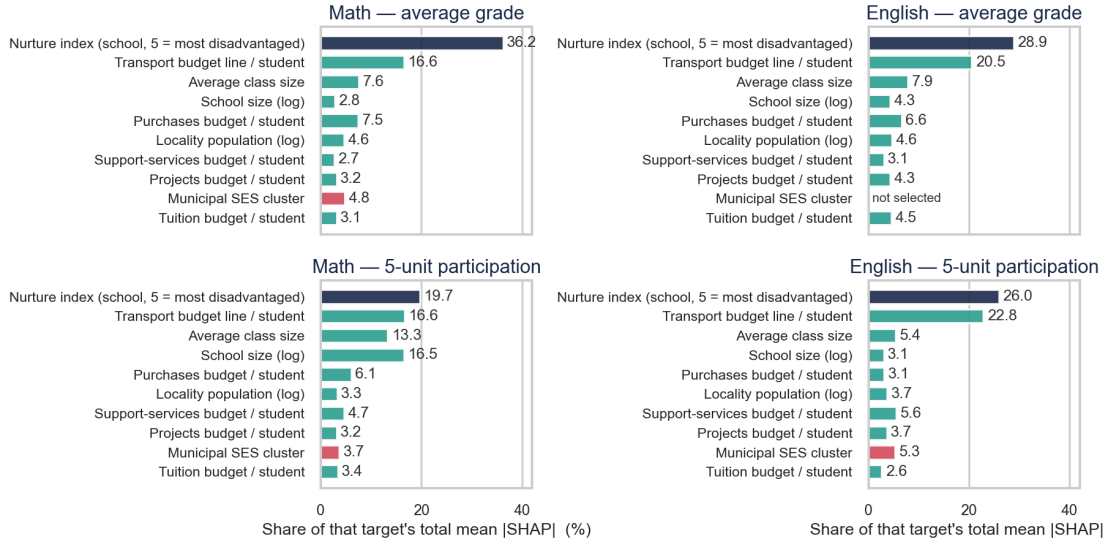


Figure 2: Mean |SHAP| importance as a share of each target’s total, from a RandomForest refit matching the selected champion. The school-level nurture index leads all four outcomes (19.7% to 36.2%), while municipal cluster never exceeds 5.3% and is not selected at all for English average grade.

How much the school-level data adds. SHAP ranks features inside a model but cannot say what the extra data was worth. So each target is modelled twice on *identical rows* under the same nested protocol: once on municipal features alone (cluster, population, locality type, year) and once on the full candidate set. Only the available information differs, so the gap measures incremental predictive value (Table 5).

Table 5: Ablation study under the same nested protocol: outer-fold R^2 (mean $\pm$ sd) from municipal features alone, and the gain from adding the school-level set on identical rows. The full-set scores are the Random Forest column of Table 3 and so are not repeated here.

Target	Rows	Municipal only	ΔR^2 from school data
Math grade	2,995	0.109 $\pm$ 0.019	+0.311 $\pm$ 0.040
English grade	2,963	0.196 $\pm$ 0.028	+0.256 $\pm$ 0.034
Math 5-unit	3,221	0.055 $\pm$ 0.033	+0.340 $\pm$ 0.039
English 5-unit	3,226	0.229 $\pm$ 0.050	+0.328 $\pm$ 0.043
Mean		0.147	+0.309

Municipal features alone predict little, with advanced Math participation close to unpredictable from the town. Adding the school-level set raises every target to between 0.395 and 0.557, a mean gain of +0.309 that holds in all 20 outer folds, so it is not an artefact of a favourable split. The result is associational: incremental predictive value, not a causal effect of school resources. The resilience question. This one uses the municipal gradient alone, holding institutional features out so the numbers read as a clean measure of socioeconomic disparity (Table 6). Comparing cluster 2 against cluster 9, Mathematics is the more resilient subject: a slightly smaller achievement gap and a participation gap roughly a third of English’s. Overachievers are clusters 1–4 schools whose average grade reaches the *median* of clusters 8–10 (84.90). Of 460 such schools, 87 (**18.9%**) clear that bar, and on Welch two-sample tests their advanced-track participation is roughly 2.7 $\times$ their peers’ in Math and 2.6 $\times$ in English.

Table 6: The resilience question, in two parts. Resilience contrasts cluster 2 with cluster 9. Overachievers are clusters 1–4 schools at or above the clusters 8–10 median grade (84.90). The comparison group is the remaining low-cluster schools.

Resilience, cluster 2 $\rightarrow$ 9	Math	English	
Grade gap (points), Cohen’s d	6.18 (0.91)	6.45 (1.16)	
5-unit participation gap	0.115	0.367	
Overachievers, 87 of 460	Over	Peers	p
Math 5-unit participation	0.132	0.049	1.4×10^{-5}
English 5-unit participation	0.468	0.181	1.4×10^{-11}

8 Results and Conclusions

The main question. Combined, municipal socioeconomics and school-level resources predict Bagrut outcomes usefully: nested out-of-fold R^2 of 0.40 to 0.56 (Table 3) with near-zero mean residuals (Figure 1). *Combined* carries it: municipal data alone reaches only 0.06 to 0.23, and the school-level features lift every target by +0.309 on average, in every outer fold (Table 5).

The resilience question. Mathematics is the more resilient subject (Table 6): the cluster 2 to cluster 9 gap is smaller for Math in achievement and about a third of English’s in participation. Overachieving schools clearly exist, 87 of 460 low-cluster schools, with significantly higher advanced-track participation than their peers.

The school-factors question. The strongest school-level predictor is the Ministry’s per-school nurture index, first for all four targets, followed by the transport-related budget field, school size, and class size. Five per-student budget ratios are confirmed for every target (Section 7). Municipal cluster never exceeds 5.3% of attribution and is dropped for English average grade.

Conclusion. Socioeconomic status matters, but the municipal average is much coarser than the school-level measure. Measured per school the same construct dominates every model. The town-level cluster is among the weakest features retained and is rejected entirely for one target.

Limitations and further work. The institutional data is a single 2014/15 snapshot applied to all four exam years, so within-school change cannot be separated. Suppressed grade cells are small cohorts, so the smallest schools are underrepresented, and complete-case modelling removes a further 9–12.5% of rows. The transport field’s negative values leave its meaning unconfirmed. Results are associational, and this model is not suitable for individual admissions decisions or for withholding resources from schools. Natural extensions: per-year budget records to test whether spending *precedes* outcomes, a study of the 87 overachievers, and bounded-response models for participation rates.

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